AI-Powered Analytics Platform for Requalification of Avionics from Recovered Vehicle Stages

Summary

- The proposal is to design and implement an AI/ML-based tool for data analytics, Avionics subsystem health assessment and requalification of a recovered launch vehicle stage. High volume of flight data to be analyzed includes sensor data and health data of recovered Avionics subsystems.
- Dataset for training and validation can be sourced from ground and flight tests of packages and subassemblies, excluding data of confidential nature. Data corresponding to subsystem faults and synthesised data from simulation test beds after fault injection can be provided as inputs.
- Pattern checking can be applied for fixed data like health bytes available in telemetry.
- Variable data like vibration, acoustics and shock from sensors shall be subjected to bound checks.
- Outliers in measurements shall be detected and conditional clearance based on decisions made during the past missions can be given with human confirmation.
- The platform shall support Extract, Transform and Load (ETL) pipelines as data processing tools for accepting, transforming and storing data from multiple sources. It shall also support image processing algorithms.
- AI/ML pipelines shall be supported to implement and improve custom analytics solutions as per specific requirements.
- Dashboarding capabilities shall be built in for visual displays in specific areas and for generating an executive overview of the recovered stage, complemented by reporting, summarizations & detailed views. Slicing & dicing features shall be provided for analysing data in multiple dimensions to gain deeper insights and aid in decision-making.
- A data warehouse capable of storing terabytes of flight data is required.

Scope of the Work

- The proposed solution shall be capable of handling high-volume data of the order of 64GB. The platform implemented shall support deterministic bound checks & ML-based solutions, complementing each other.
- Focus areas are requalification of flight-deployed Avionics systems for severe damage, performance deviations, and minor anomalies. This shall enable fast and accurate recertification of subsystems for reuse. It shall also minimize manual inspection time and human errors.
- Assessments will have to be integrated to subsequent launch vehicle missions also, on a continuous operational basis. For this, data shall be fed forward from previous simulations, flights etc.

Linkages to Space Programme

- The platform can be deployed for analysis of flight and health data from past missions.
- It can aid fast and accurate decision-making regarding reuse of recovered Avionics subsystems (as planned for future missions like NGLV) with minimal refurbishment, significantly reducing launch costs and increasing mission frequency.

Expected Deliverables

- AI-Powered Analytics Platform – source code, executables, lifecycle documentation.
- Platform deployed internally at VSSC, in a server cluster, with required computing, communication & storage capability.
- Demonstration of data warehouse, ETL pipeline & AI/ML pipeline.